

Practice Questions for 02L2-Intro

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December 30, 2024

1 Questions

1. According to the lecture, what was a key goal of the 1956 Dartmouth summer research project?
 - a. To develop a computer that could play chess at a grandmaster level.
 - b. To create a machine that could simulate any aspect of learning or intelligence.
 - c. To build a robot capable of performing physical tasks.
 - d. To establish a new programming language for AI research.
 - e. To create a database of all human knowledge.
2. Which of the following best describes the Turing Test?
 - a. A test to determine if a machine can perform complex mathematical calculations.
 - b. A test to determine if a machine can exhibit intelligent behavior indistinguishable from a human.
 - c. A test to determine if a machine can learn from a large dataset.
 - d. A test to determine if a machine can physically interact with the environment.
 - e. A test to determine if a machine can generate creative art.
3. According to the lecture, why is search alone insufficient for complex games like Checkers or Go?
 - a. Because the rules of these games are too complex to encode.
 - b. Because the search space is too vast, making it computationally infeasible to explore all possibilities.
 - c. Because these games require human-like intuition and creativity.
 - d. Because computers are not fast enough to perform the necessary calculations.
 - e. Because these games require a large amount of memory to store the game states.
4. What is a key difference between how Deep Blue and AlphaGo achieved success in their respective games?
 - a. Deep Blue relied heavily on machine learning, while AlphaGo used brute-force search.
 - b. Deep Blue used a combination of deep learning and search, while AlphaGo relied on human intuition.

- c. Deep Blue's success was largely due to brute-force search, while AlphaGo used a combination of deep learning and search.
 - d. Deep Blue relied on a single model, while AlphaGo used two models.
 - e. Deep Blue was trained on self-play, while AlphaGo was trained on human games.
- 5.
- a. According to the lecture, what is a key aspect of the 1956 Dartmouth summer research project?
 - A. To develop a machine that can play chess at a grandmaster level
 - B. To explore the possibility of simulating any aspect of learning or intelligence with a machine
 - C. To create a computer program that can pass the Turing Test
 - D. To build a robot that can perform physical tasks
 - E. To develop a new programming language for AI
 - b. Which of the following best describes the Turing Test?
 - A. A test to measure a machine's ability to perform complex calculations
 - B. A test to determine if a machine can exhibit intelligent behavior indistinguishable from a human
 - C. A test to evaluate a machine's physical capabilities
 - D. A test to measure a machine's memory capacity
 - E. A test to assess a machine's ability to learn from data
 - c. According to the lecture, why is search alone insufficient for complex games like Checkers or Go?
 - A. Because the rules of these games are too complex to encode
 - B. Because the search space is too large to explore exhaustively
 - C. Because these games require human intuition and creativity
 - D. Because computers are not fast enough to perform the necessary calculations
 - E. Because these games require real-time decision-making
 - d. What is a key difference between how Deep Blue and AlphaGo achieved success in their respective games?
 - A. Deep Blue relied primarily on machine learning, while AlphaGo used brute-force search
 - B. Deep Blue used a combination of machine learning and search, while AlphaGo relied solely on search
 - C. Deep Blue relied primarily on brute-force search, while AlphaGo used a combination of deep learning and search
 - D. Deep Blue used a rule-based system, while AlphaGo used a neural network
 - E. Deep Blue used a Monte Carlo search, while AlphaGo used a minimax algorithm
 - e. According to the lecture, what is a key component of an algorithm for playing games?
 - A. The ability to generate random moves
 - B. The ability to represent the state of the game
 - C. The ability to communicate with human players
 - D. The ability to access external databases

- E. The ability to predict the weather
6. According to the lecture, what is the primary goal of the initial lab assignment?
 - a. To write a complex AI algorithm from scratch
 - b. To explore and experiment with existing AI tools
 - c. To create a detailed report on the history of AI
 - d. To prepare for an in-person lab session
 - e. To solve a specific problem using AI
 7. According to the lecture, what is the main focus of the lecture?
 - a. Advanced machine learning techniques
 - b. The history of AI, the nature of intelligence, and characteristics of the field
 - c. Specific AI algorithms for game playing
 - d. Ethical considerations in AI development
 - e. The mathematical foundations of AI
 8. According to the lecture, what is the core idea behind the Turing Test?
 - a. To measure a machine's ability to perform complex calculations
 - b. To determine if a machine can exhibit intelligent behavior indistinguishable from a human
 - c. To assess a machine's ability to learn from large datasets
 - d. To evaluate a machine's physical capabilities
 - e. To test a machine's ability to follow a set of instructions
 9. According to the lecture, what is a key limitation of using search alone for complex games?
 - a. Search algorithms are too slow to be practical
 - b. Search algorithms cannot handle games with randomness
 - c. The search space for complex games is too large to explore exhaustively
 - d. Search algorithms cannot learn from past experiences
 - e. Search algorithms are not compatible with modern hardware
 10. According to the lecture, what is a key component of an algorithm for playing games?
 - a. A representation of the game state
 - b. A complex neural network
 - c. A large dataset of human games
 - d. A powerful graphics processing unit
 - e. A detailed simulation of the real world

2 Answers

1. According to the lecture, what was a key goal of the 1956 Dartmouth summer research project?
 - a. While game playing is a part of AI, the Dartmouth project's goal was broader than just chess.
 - b. The Dartmouth project aimed to explore the possibility of machines simulating any aspect of learning or intelligence, as stated in the proposal quote on the slide.
 - c. The Dartmouth project focused on cognitive abilities, not physical tasks.
 - d. The project's goal was not to create a new programming language.
 - e. The project's goal was not to create a database of all human knowledge.
2. Which of the following best describes the Turing Test?
 - a. The Turing Test is not about mathematical calculations.
 - b. The Turing Test is designed to assess a machine's ability to exhibit intelligent behavior that is indistinguishable from that of a human.
 - c. The Turing Test is not about learning from data, but about exhibiting intelligent behavior.
 - d. The Turing Test is not about physical interaction.
 - e. The Turing Test is not about creative art.
3. According to the lecture, why is search alone insufficient for complex games like Checkers or Go?
 - a. The rules of these games are well-defined and can be encoded.
 - b. The lecture states that the search space for games like Checkers and Go is so large that it would take an impractical amount of time to search the entire game tree, making brute-force search impossible.
 - c. While intuition and creativity are important, the main issue is the size of the search space.
 - d. While computational speed is a factor, the main issue is the exponential growth of the search space.
 - e. While memory is a factor, the main issue is the exponential growth of the search space.
4. What is a key difference between how Deep Blue and AlphaGo achieved success in their respective games?
 - a. Deep Blue did not rely heavily on machine learning, and AlphaGo did not use brute-force search.
 - b. Deep Blue did not use deep learning, and AlphaGo did not rely on human intuition.
 - c. The lecture states that Deep Blue's success was largely due to brute-force search, while AlphaGo used a combination of deep learning and search.
 - d. Both Deep Blue and AlphaGo used a single model.
 - e. AlphaGo was trained on both human games and self-play.
5.
 - a. According to the lecture, what is a key aspect of the 1956 Dartmouth summer research project?

- A. Answer A is incorrect because while game playing is a part of AI, the Dartmouth project's goal was broader than just chess.
 - B. Answer B is correct because the Dartmouth project aimed to explore the simulation of learning and intelligence in machines, as stated in the proposal.
 - C. Answer C is incorrect because the Turing Test was not the primary focus of the Dartmouth project.
 - D. Answer D is incorrect because the Dartmouth project focused on cognitive simulation, not physical robotics.
 - E. Answer E is incorrect because the Dartmouth project did not focus on developing a new programming language.
- b. Which of the following best describes the Turing Test?
- A. Answer A is incorrect because the Turing Test is not about computational ability, but about mimicking human-like intelligence.
 - B. Answer B is correct because the Turing Test is designed to assess a machine's ability to exhibit intelligent behavior that is indistinguishable from a human.
 - C. Answer C is incorrect because the Turing Test is not about physical capabilities, but about cognitive abilities.
 - D. Answer D is incorrect because the Turing Test is not about memory capacity, but about intelligent behavior.
 - E. Answer E is incorrect because the Turing Test is not specifically about learning, but about the overall appearance of intelligence.
- c. According to the lecture, why is search alone insufficient for complex games like Checkers or Go?
- A. Answer A is incorrect because the rules of these games are well-defined and can be encoded.
 - B. Answer B is correct because the lecture emphasizes that the search space for complex games is too large to explore exhaustively, making brute-force search infeasible.
 - C. Answer C is incorrect because while human intuition is valuable, AI can learn to play these games effectively without it.
 - D. Answer D is incorrect because while computational speed is a factor, the primary issue is the exponential growth of the search space.
 - E. Answer E is incorrect because while real-time decision-making is important, the primary issue is the size of the search space.
- d. What is a key difference between how Deep Blue and AlphaGo achieved success in their respective games?
- A. Answer A is incorrect because Deep Blue relied on brute-force search, not machine learning.
 - B. Answer B is incorrect because AlphaGo used a combination of deep learning and search, not solely search.
 - C. Answer C is correct because Deep Blue relied heavily on brute-force search, while AlphaGo used a combination of deep learning and Monte Carlo tree search.
 - D. Answer D is incorrect because Deep Blue did not use a rule-based system, but rather brute-force search.
 - E. Answer E is incorrect because Deep Blue did not use a Monte Carlo search, and AlphaGo did not use a minimax algorithm.

- e. According to the lecture, what is a key component of an algorithm for playing games?
 - A. Answer A is incorrect because while randomness can be a part of some algorithms, it is not a key component for all game-playing algorithms.
 - B. Answer B is correct because the lecture emphasizes that representing the state of the game is a key component of any game-playing algorithm.
 - C. Answer C is incorrect because communication with human players is not a necessary component of a game-playing algorithm.
 - D. Answer D is incorrect because accessing external databases is not a key component of a game-playing algorithm.
 - E. Answer E is incorrect because predicting the weather is not relevant to game-playing algorithms.
- 6. According to the lecture, what is the primary goal of the initial lab assignment?
 - a. Answer A is incorrect because the lab is not about writing algorithms from scratch, but rather exploring existing tools.
 - b. Answer B is correct because the slide states that the initial lab is to "Play with AI" and is not in person.
 - c. Answer C is incorrect because the lab is not about the history of AI, but rather about using AI tools.
 - d. Answer D is incorrect because the lab is not in person.
 - e. Answer E is incorrect because the lab is not about solving a specific problem, but rather about exploring AI tools.
- 7. According to the lecture, what is the main focus of the lecture?
 - a. Answer A is incorrect because the lecture is an introduction to AI, not advanced techniques.
 - b. Answer B is correct because the slide explicitly lists "History", "The nature of intelligence", and "Characteristics of the field of study" as the topics for the lecture.
 - c. Answer C is incorrect because the lecture is an introduction to AI, not specific algorithms.
 - d. Answer D is incorrect because the lecture is an introduction to AI, not ethical considerations.
 - e. Answer E is incorrect because the lecture is an introduction to AI, not mathematical foundations.
- 8. According to the lecture, what is the core idea behind the Turing Test?
 - a. Answer A is incorrect because the Turing Test is not about complex calculations, but rather about intelligent behavior.
 - b. Answer B is correct because the Turing Test is designed to see if a machine can behave intelligently, indistinguishable from a human.
 - c. Answer C is incorrect because the Turing Test is not about learning from datasets, but rather about exhibiting intelligent behavior.
 - d. Answer D is incorrect because the Turing Test is not about physical capabilities, but rather about intelligent behavior.
 - e. Answer E is incorrect because the Turing Test is not about following instructions, but rather about exhibiting intelligent behavior.

9. According to the lecture, what is a key limitation of using search alone for complex games?
- a. Answer A is incorrect because while search can be slow, the main issue is the size of the search space.
 - b. Answer B is incorrect because search algorithms can handle randomness, but the main issue is the size of the search space.
 - c. Answer C is correct because the slide explicitly states that the search space for games like checkers is too large to search exhaustively, and that "Search alone is impossible, we need machine learning."
 - d. Answer D is incorrect because while search algorithms may not learn from past experiences, the main issue is the size of the search space.
 - e. Answer E is incorrect because search algorithms are compatible with modern hardware, but the main issue is the size of the search space.
10. According to the lecture, what is a key component of an algorithm for playing games?
- a. Answer A is correct because the lecture specifically mentions that a representation of the game state is a key component of an algorithm for playing games.
 - b. Answer B is incorrect because while neural networks can be used in game playing, they are not a key component of all game playing algorithms.
 - c. Answer C is incorrect because while datasets can be used to train game playing algorithms, they are not a key component of all game playing algorithms.
 - d. Answer D is incorrect because while GPUs can be used to accelerate game playing algorithms, they are not a key component of all game playing algorithms.
 - e. Answer E is incorrect because a detailed simulation of the real world is not necessary for all game playing algorithms.